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Abstract

The purpose of this report is to provide a comprehensive analysis of Business Continuity Planning and Disaster Recovery Planning, while reinforcing the necessity of developing resilient organizations following disruptive events. Business continuity plan is an activity that maintains the viability of business by preserving the critical functions, processes, and services impacted by disruptive events. Disaster recovery is the process used to recover the technology solutions, data, and IT infrastructure that aid in facilitating business operations; as such, both frameworks share an inter-relationship.

It's important to note that disaster recovery is not only a subset of business continuity plan, but that disaster recovery must be integrated with the business continuity goals. The report compares additional forms of risk management including ISO 22301 business continuity standard and NIST disaster recovery guidelines, in order to provide organizations with a structured approach to planning, implementing, testing, monitoring, and continuously improving the organization's ability to provide business continuity and disaster recovery measures.

By integrating business continuity and disaster recovery plans, organizations will be able to minimize downtime, reduce financial and operational losses, protect critical data and infrastructure, maintain customer confidence, and sustain business operations. As businesses strive long-term viability, growth and competitiveness in today's hyperconnected world, a soundly developed integrated strategic approach to continuity and recovery will play an increasingly important role in their ultimate success.

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1. Introduction

1.1 Background

This report analyzes about BCP (Business Continuity Plan) and DRP (Disaster Recovery Plan), which are very essential components while building a modern organization. These plans support them during crucial disruptive times by reducing the potential loss and helping them prepare for unexpected events. Real-life cases are also discussed further in this report, which shows how the companies plan, act and recover from these kinds of problems; how they learned from their experience; and how they improved their strategies (team T. o., 2024).

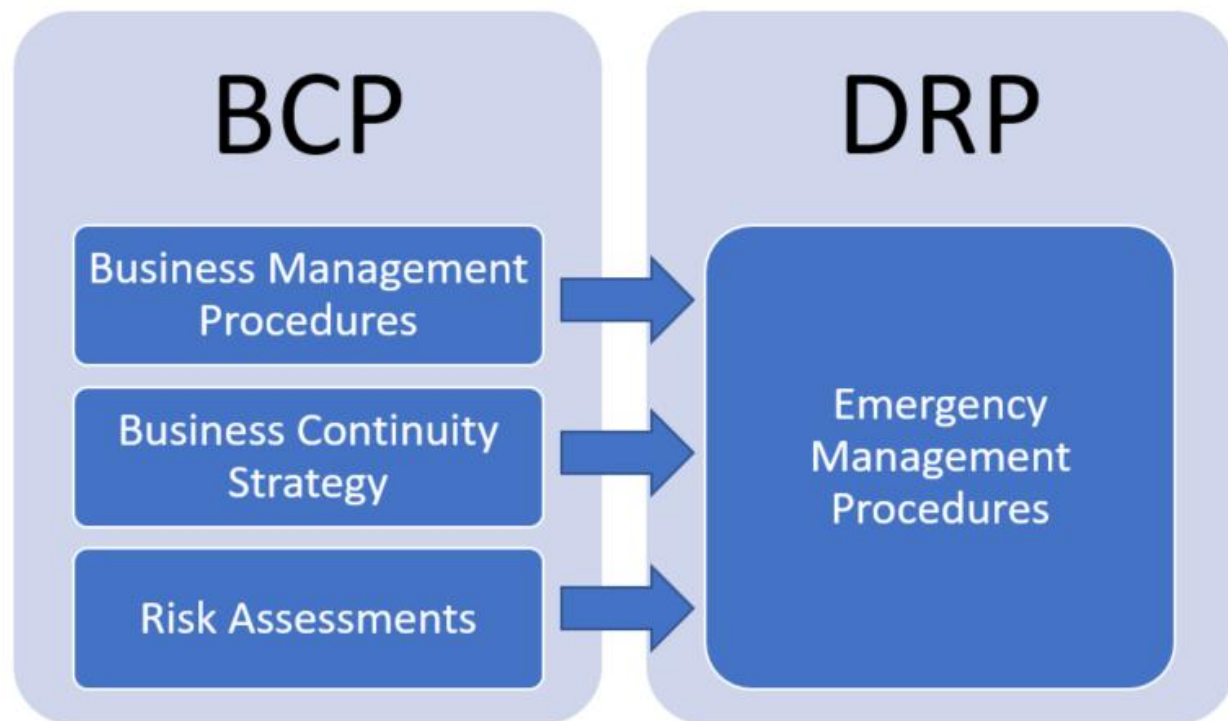


Figure 1: BCP and DRP (Larkins, 2018).

Risk identification and analysis are the key of BCP and DRP because if we can identify it, we can analyse, plan, and test our strategies, which helps protect both the reputation and financial stability of the organization (team P. , 2025).

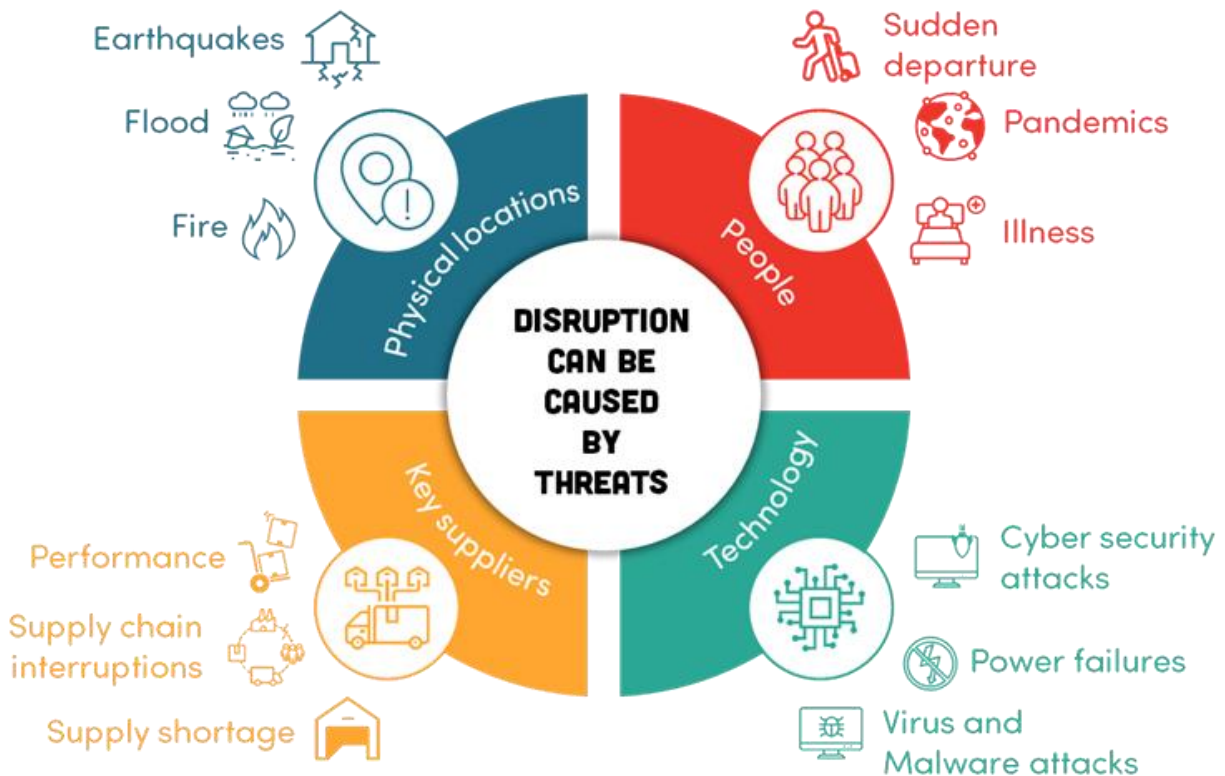


Figure 2: Possible Risk and Threats (Transformation, 2025).

Studies show that more than 70% of companies/organizations face at least one impactful business disruption every year (organization, 2023). IBM has reported according to 2024 that the global average cost of data breaches reached \$4.88 million which was 10% more than the prior year. Also, that implementation of BCP and DRP strategies can reduce the recovery time by up to 50%. (CAMBRIDGE, 2024). In 2020, during the unexpected worldwide health crisis Covid-19 pandemic, business continuity statistics show that around 700,000 business were closed in the second quarter (some of them reopened after some time) (Shulmistra, 2025).

1.2 Rationale

The rationale of this report is to understand how BCP and DRP works and how companies applied it practically in real-world scenarios—learning what worked and what failed for them. By examining case studies, it can be concluded how planning beforehand has so much importance for an organization. A strong and good Business Continuity Plan can reduce the possibility of risk during a moment of crisis and Disaster Recovery Plan restores critical data after incidents of crisis. Comprehending real-world cases helps in better decision-making during emergencies. The uncovering of this report can help and guide organizations in thinking up for future emergencies.

1.3 Objectives

The objectives for this report are:

- To understand Business Continuity Plan (BCP) and Disaster Recovery Plan (DRP) through real life cases.
- To identify what makes a BCP and DRP fortunate during crisis.
- To comprehend the challenges and dangers faced by the organizations in a time of disaster.
- To inspect how the organizations responded and acted during unexpected event occurrence.
- To estimate how swiftly and effectively organizations rehabilitated after the disruption.

2. Literature Review

2.1 Business Continuity Plan (BCP)

A Business Continuity Plan (BCP) is a structured strategies used by organizations to make sure that the essential and most critical operations carry on with during and after disruptions and disasters. This plan generally involves procedure to ensure the security of the critical operations carried on by the organization, and mitigate financial and reputational loss during the time of disruption, focusing on quickly recover and reinstitute normal organization activities (Pohlmann, 2023).

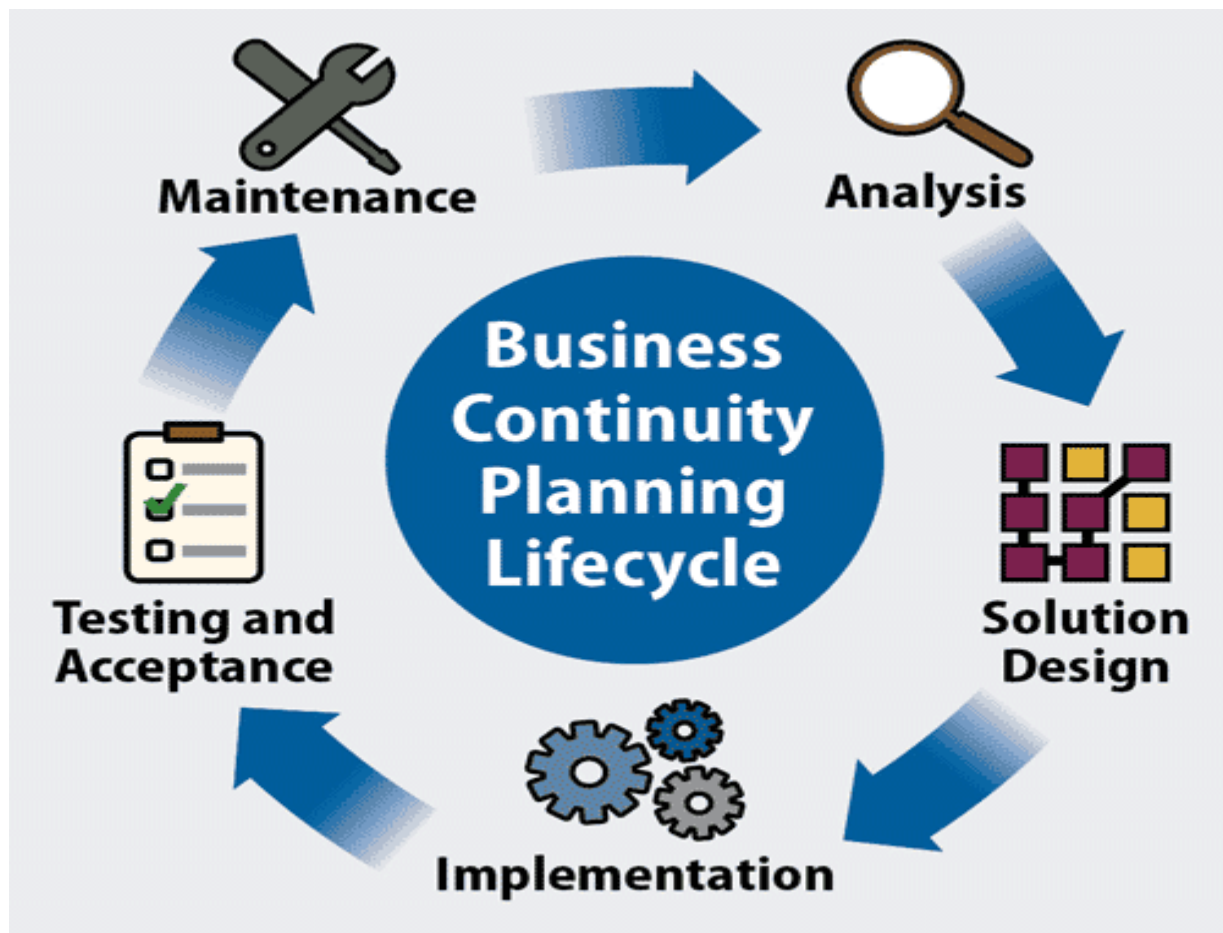


Figure 3: BCP framework (Emigh, 2019).

The core components of a practical and functional Business Continuity Plan are Risk Identification and Assessment, Business Impact Analysis and Recovery Strategies. Risk Assessment is a structured procedure of recognizing potential risks that could disrupt organization operations. Business Impact Analysis is a mechanism used to acknowledge the outcome of disruption on critical organization operations. Recovery Strategies are the plans formed based on the understandings learned from risk assessment and business impact analysis (LAW, 2025).

2.1.1 ISO 22301

ISO 22301 is the international standard for Business Continuity Management Systems (BCMS). It hands over a structured framework to plan, establish, implement, operate, monitor, review and maintain, which helps organizations be prepared for any kind of disruption. This standard is essential for businesses to identify risks, prepare for emergencies, and improve recovery time. Enhancing organizational flexibility, improving risk management, systematic response to the disruption and Increase trust are some of the benefits of ISO 22301 (ISO, 2019).

2.2 Disaster Recovery Plan (DRP)

A Disaster Recovery Plan (DRP) is a comprehensive document that outlines how an organization will acknowledge productively to a spontaneous and unexpected occurrence and carry on with the business functions. DRP helps an organization to make sure that they are ready and well prepared to tackle with various types of disasters and disruptions including power outages, malwares, natural disasters and many others. A secure and stable DRP efficiently and smoothly helps re-establish connectivity and patching the data loss after a disruption (Stephanie Susnjara, 2023).

Once an organization has pinpointed their restoration objectives, then they can choose and implement the proper disaster recovery plans. The Benefits of DRP are:

- Recovering the reputational damage and financial losses.
- Helps organizations respond in time and building up customer trust.
- DRP is very crucial in today's industry to keep up with every other organization and be their competition (Carey, 2023).

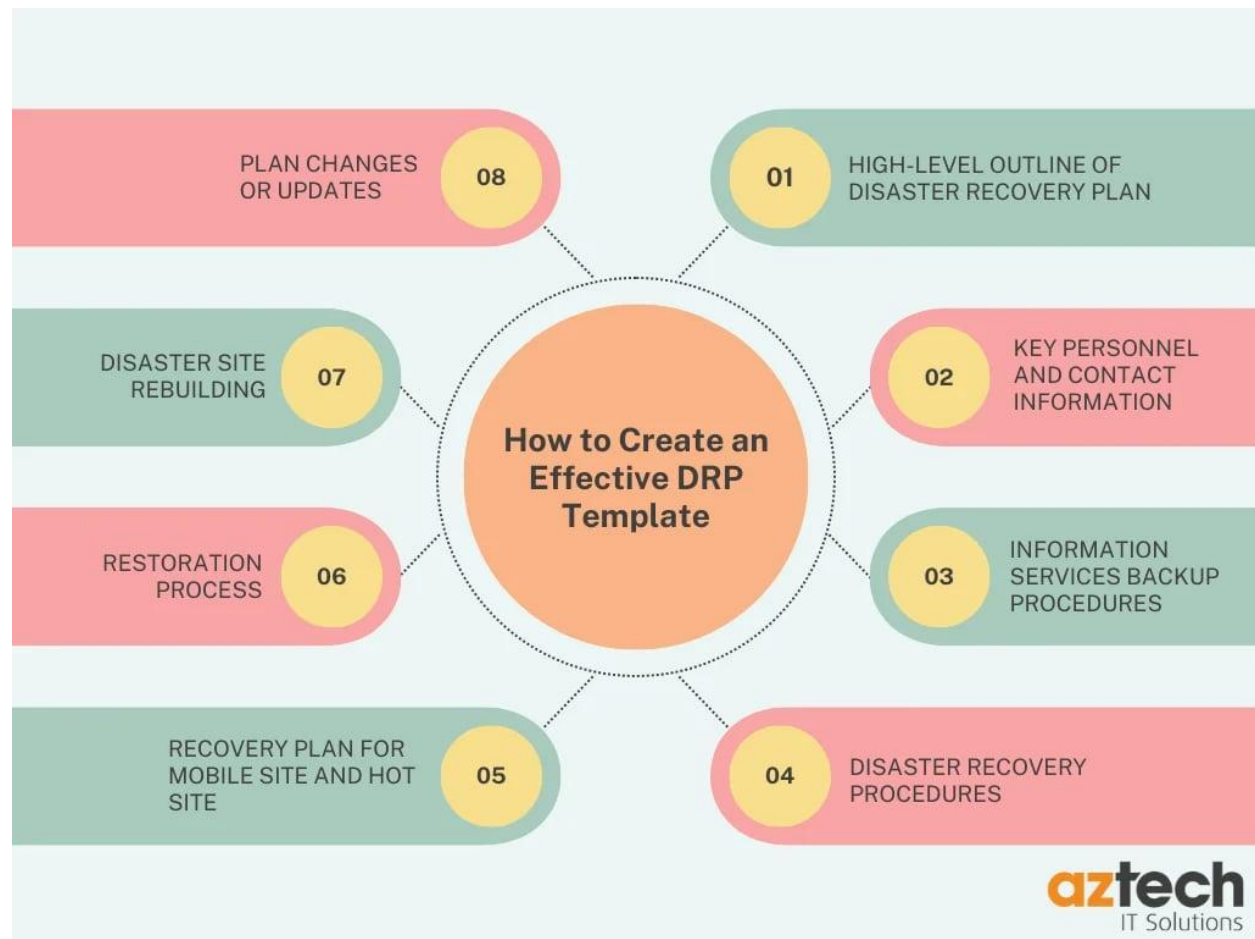


Figure 4: DRP framework (team A. , 2023).

The National Institute of Standards and Technology (NIST) provides strong and proper guidance for disaster recovery planning. It mainly focuses on information systems and IT infrastructures and helps businesses to identify critical data. Supports to emphasize risk assessment and business impact analysis. It also suggests that systematic testing, and revising the plan regularly is very essential (Marianne Swanson, 2010).

2.3 Relation Between BCP and DRP

BCP makes sure that a business operation continues to run without any problem and interruptions during crisis while DRP are created to restore IT Systems and Data which supports business operations. As a result, both BCPs and DRPs play critical roles in maintaining and supporting an organization's longevity (Abdelhameed, 2024).



Figure 5: Relation Between BCP and DRP (Geeksforgeeks, 2025)

2.4 Literature Review Summary

According to the literature review, business continuity planning (BCP) and disaster recovery planning (DRP) are vital for keeping organizations operating after a disruption and are intimately related. The BCP focuses on assessing risks and the impact on an organization's ability to function during a disruption, while the DRP addresses the restoration of IT systems, data, and infrastructure necessary to support the operation of the business after the disruption occurs. BCP and DRP utilize internationally known frameworks (ISO 22301 and National Institute of Standards and Technology (NIST) consulting practices) as a defined method for the planning of, testing of, and continual improvement of these processes. BCP plus DRP minimizes a company's downtime, reduces losses (financial and reputational), and builds trust in the company by its customers. The literature review makes clear that BCP and DRP must be integrated for the survival of the modern organization due to organizations becoming more dependent on IT to conduct business processes.

3. Purposed Case Studies

3.1 Case 1: Maersk – NotPetya Cyberattack (2017)

3.1.1 Background

During 2016, A.P. Moller – Maersk, better known as Maersk, was one of the biggest and no.1 shipping company by volume. Globally it had over 75,000 employees in 130 countries overseeing logistics, ports, and shipping lines and was considered one of Denmark's largest companies (University, 2022).

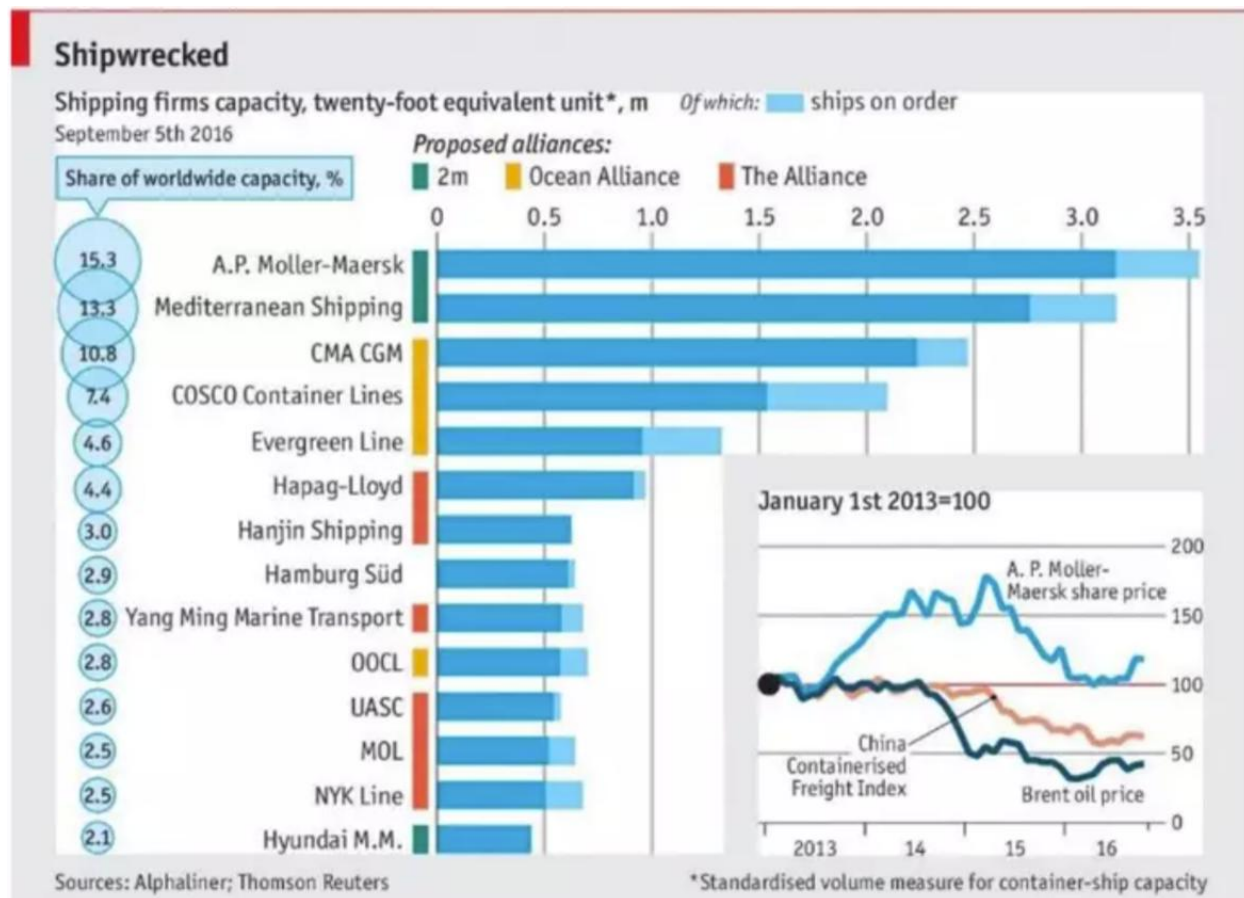


Figure 6: A.P. Moller - Maersk (Pownall).

3.1.2 Description of the Attack

In June 2017, Maersk fell victim to the NotPetya malware, a sophisticated cyberattack which was believed to be state-sponsored. This malware which was camouflaged as ransomware spread swiftly across the company's global IT infrastructure, Weakening terminal operations, surveillance system, and internal communications.

After this attack the company lost access to almost about 49,000 laptops and over 1,000 applications. Ports throughout the globe were left without ability at once to assist with the shipment of cargo. Although Maersk did not directly suffer from an attack, they experienced significant loss due to their position as one of the largest shipping companies in the world. The malware originally propagating through Ukrainian accounting programs was able to utilize Maersk's worldwide network system and spread rapidly on both an extensive scale and within a very short time frame (YALLO).

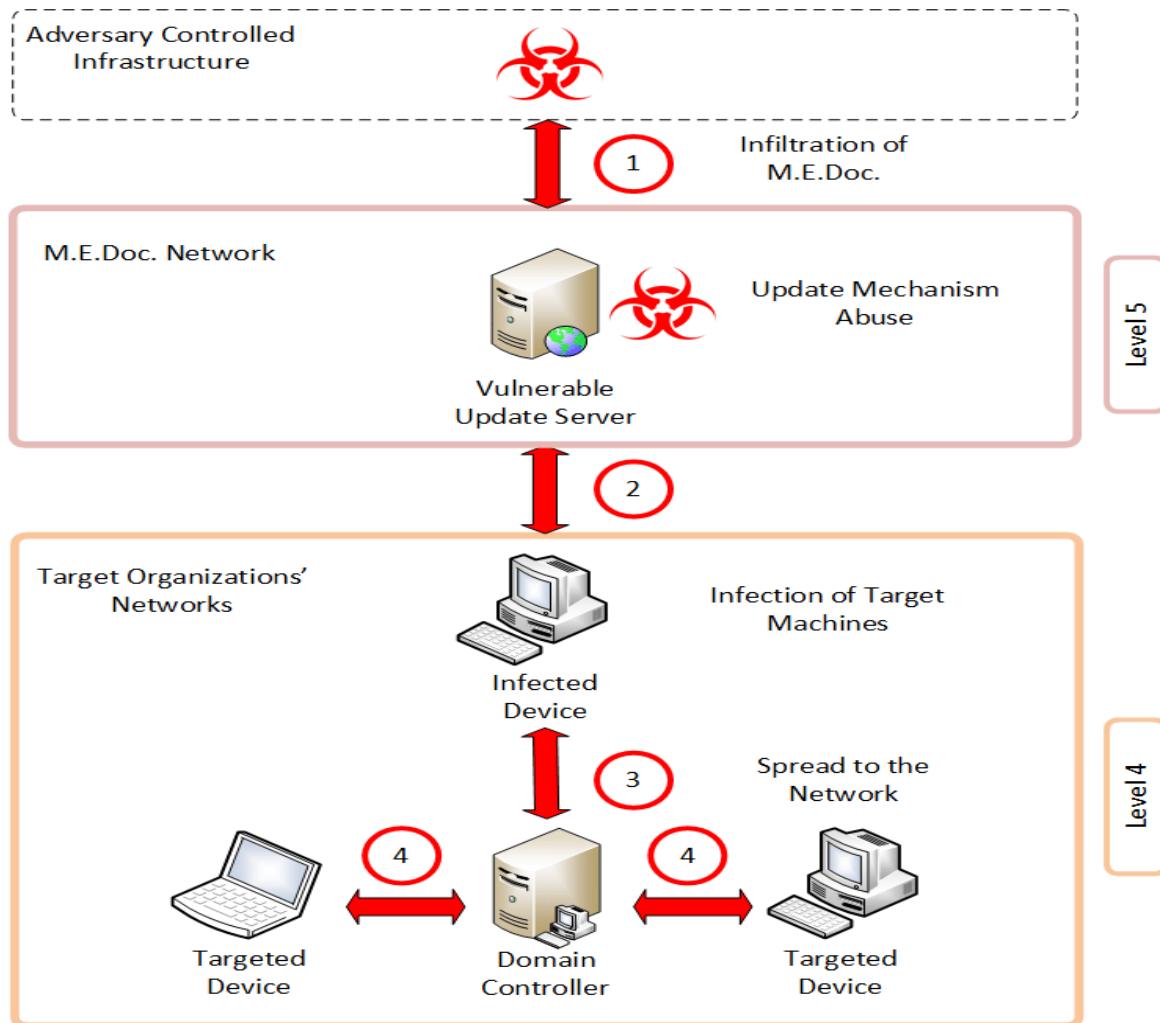


Figure 7: The NotPetya Attack Process (Georgios Michail Makrakis, 2021).

3.1.3 Financial and Operational Impact

The consequences of this attack were astonishing, the company approximately lost US\$300 million due to project delays, port closures, and cost incurred during recovery, delay and divert in marine transportation for multiple vessels, causing disruptions to global supply chains and thousands of affected shipments resulted in contract penalties.

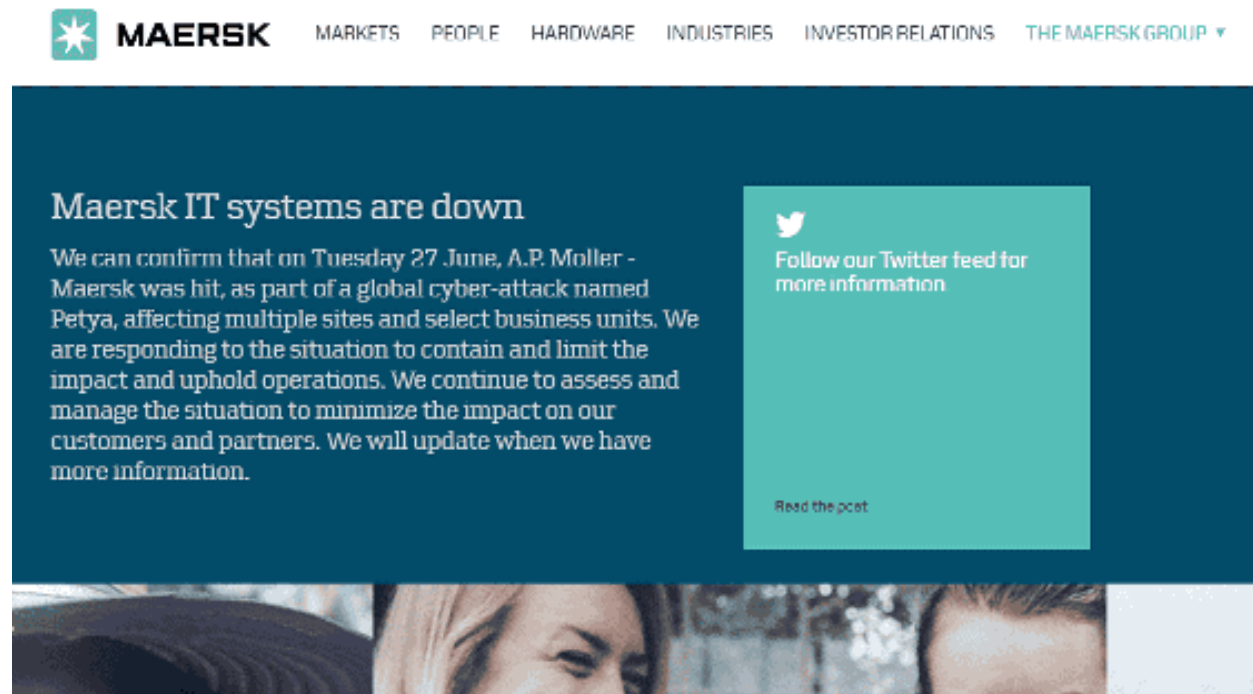


Figure 8: Screenshot of the website during the attack (Gigazine, 2017).

Maersk had rebuilt its complete IT infrastructure with in about 10 days to restore its business, it was able to do so because there was a copy of the Active Directory on site at an office that was not affected by the attack. This turned out to be a valuable resource for Maersk, allowing them to quickly resume operations and avoid a long period of operational downtime.

3.1.4 Response Strategies

As part of its response to the seriousness of the breach, Maersk set up a Cross-Functional Data Risk and Governance Office to take an enterprise-wide view. This included doing an end-to-end data lineage map of its environments to determine how data flowed through its systems so it could identify the paths where malware had the potential to attack or compromise them. They performed "data blast radius" assessments to understand the extent to which malware could propagate within their system, including how far down the architecture it might spread. As a result of these findings, Maersk established a global backup/recovery program for rebuilding its IT infrastructure with high levels of segmenting and redundancy. In addition, Maersk formalized its incident response protocols by creating detailed playbooks for its crisis management processes to ensure that it was prepared for future attacks.

3.1.5 Summary

This case study explores the impact of a Russian computer attack against Ukraine and other businesses (commonly referred to as "NotPetya") on the Danish shipping company A.P. Moller-Maersk. Maersk is not the only large organization targeted within this Russian cyber operation. The focus of this case study is on part the response of Maersk during the period when their computer systems were compromised. It demonstrates the role of components of Maersk's Cybersecurity Program in limiting the spread of NotPetya malware and the effect that it had on the business operations of Maersk for days after the attack had taken place. Among other issues, this case study provides an understanding of the relationship between network segmentation and an effective data recovery plan.

3.2 Case 2: British Airways – IT System Failure (2017)

[Appendix Case Study: British Airways](#)

4. Conclusion

In today's world, to be effective and maintain control of your business during an unexpected event, you will want to create a thorough Business Continuity Plan (BCP) and Disaster Recovery Plan (DRP). You may want to look closely at the incidents such as the Maersk ransomware attack and the British Airways IT system failure, which demonstrate how disruptions to IT systems cause significant financial losses, delays in operation, and loss of credibility.

The Maersk example reinforces the necessity of having a strategy for how to protect your IT systems from cyberattacks and how you will recover quickly from a serious disruption to your IT systems. The example of British Airways emphasizes the need for reliable infrastructure, backup systems, and contingency planning to avoid chaos when an IT system is disrupted.

In both examples, the BCP and DRP are not simply technical requirements; they are vital to the business and the ability of that business to continue operations, serve customers, and recover quickly from a crisis. You should continually test and update your BCP and DRP to address cyber threats and address IT system failures.

In conclusion, BCP and DRP help reduce risk, minimize loss, and help create a resilient organization in a business world that relies on technology and is increasingly interconnected.

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5. Appendix Case Study: British Airways

5.1 Background

British Airways (BA) is the official airline of the UK and was created from the merger of British Overseas Airways Corporation (BOAC) and British European Airways (BEA) on March 31, 1974. It is located at Waterside near London Heathrow Airport and belongs to the International Airlines Group, which operates as the world's largest airline alliance according to total revenue passenger kilometres (RPK). The BA fleet consists of over 250 commercial jets serving over 180 international routes; cargo and mail are also transported on these planes. BA offers a loyalty club for its frequent fliers, known as the Executive Club, which offers members the chance to earn Avios points. With about 42,000 employees, British Airways provides high-quality service to its customers and has a strong focus on overseas travel and connectivity.

5.2 Description of the Attack

An IT systems outage occurred on May 27, 2017, at British Airways and caused problems for customers on a global scale. Services for the company that are impacted by the outage include, online check-in and baggage handling, as well as booking services and airport check-in counters. As a result, many passengers were left stranded, with hundreds of flights cancelled from the major airports in Heathrow and Gatwick, and severe delays were incurred. The outage was a result of a power supply to the airline's main data centre, which in turn caused server failures and eventual failure of the backup system. While there was no evidence that a cyberattack had taken place, this incident highlighted the dependence that British Airways has on its IT infrastructure, as well as the requirement to strengthen the needs for contingency and disaster recovery planning (CNCB, 2017).

5.3 Financial and Operational Impact

The computer systems failure caused severe financial and operational effects on the operations of British Airways. Operationally, because of this failure, British Airways were unable to operate any flights from their two main UK terminals, London Heathrow and London Gatwick, on the first day of the incident, and many more aircraft were to be taken off later that weekend and were cancelled or delayed due to the outage. Due to this outage, around 75,000 people were affected, leading to long wait times in airports and resulting in major disruptions around a busy holiday period. The outage also

caused significant disruption in the handling of passenger bags by causing staff not to be able to access systems to process luggage in a timely manner. As a result of this incident, analysts believe that the overall impact that British Airways incurred because of this outage was exceed £80 million to £100 million in costs related to customer compensation, refunds, and rebooking, along with lost revenue and increased expenses associated with providing customer service. One day's loss was approximately £30 million in terms of lost revenue to the airline. This incident also caused a decline in the stock price of British Airways' parent company. The prolonged outage and the resultant liabilities associated with compensating customers show the significant financial costs associated with inadequate contingency planning for IT failures in a large operational network (Johnston, 2017).

5.4 Response Strategies

To handle the IT systems failure, British Airways implemented immediate measures to respond to the crisis and assist customers impacted. British Airways cancelled all flights from Heathrow and Gatwick airports on Leek the first day of the outage, to mitigate the risk of additional operational chaos. The airline took to media and social media to offer public apologies for the inconvenience caused to passengers. BA also introduced a rebooking policy with flexible terms for passengers whose travel had been disrupted, as well as offering complete refunds to passengers whose travel had been interrupted. BA added more staff to airports and call centres to handle queues and assisted the passengers by supplying information and processing claims. In addition, the airline engaged an independent firm of investigators to determine the root cause of the outage so that similar situations would not occur again. Although no evidence of a cyberattack was presented, the airline used the recovery process to also review their backup systems, power supply protocols, and overall IT infrastructure resiliency (Johnston, 2017).

5.5 Summary

The reason that British Airways experienced an IT systems failure in May 2017 was due to an issue with the power supply at their main data centre. All servers and backup systems were affected by this disaster. Consequently, British Airways cancelled and delayed hundreds of flights, impacting around 75,000 people, significantly disrupting operations at both Heathrow and Gatwick airports. The airline estimated that the total financial impact for them would fall between £80 million and £100 million because of compensating passengers, refunding customers, the cost of rebooking customers, lost revenues, etc. The incident highlighted the need for all airlines to invest in IT infrastructure, Business Continuity Planning (BCP) and Disaster Recovery Planning

(DRP) strategies to ensure that their operations run smoothly. The company cancelled several flights, assigned additional staff, and re-evaluated the functionality of their IT infrastructure and contingency plans in reaction to the event. This example demonstrates that even large, established companies can incur tremendous operational and financial losses from unforeseeable IT catastrophes, which underscores the necessity of substantial BCP and DRP measures.